

ACTIVITY 1 — FACILITATOR GUIDE

Define — Customer Requirements and the Problem Statement

TYPE	DURATION	MAPS TO	TOPIC
Hands-on Lab	30 minutes	DEFINE	Define — Understand the Problem

Purpose of this activity

By the end of this activity learners will be able to identify customer requirements and write a clear problem statement (A1, A2).

What learners produce

A VOC-to-CTQ table with five measurable requirements, a written problem statement containing no solution, a SMART goal, and an in/out scope table.

Materials required

- Learner Worksheet (one per learner)
- Checklist (one per group)
- Flipchart or A3 sheet and markers
- The course slides for the scenario and framework
- Internet access where the activity calls for live research

Set-up

Groups of 3–5. Distribute one worksheet per learner and one checklist per group, and make sure every group can open this activity's data/ folder. All 5 activities run on the same continuous BrewBean Cafe scenario and follow DMAIC in order, so remind learners where this activity sits in that story — and that its output feeds the next one.

Scenario / briefing material

BrewBean Cafe is a busy coffee shop in the CBD serving mainly office workers on their way to work. Over the past two months online complaints about slow service have been rising, and several regulars have said they now buy their coffee elsewhere. Management has formed a small improvement team and you have joined it as the White Belt member, supporting the team rather than leading the project.

Nobody in the cafe agrees on what the problem actually is. The owner thinks the team needs a second coffee machine. The morning barista says the queue is fine and customers are simply impatient. One of the baristas is convinced the real issue is that orders keep coming out wrong. No one has asked the customers, and no one has written the problem down.

Before the team spends a dollar, your job is to find out what customers actually want, turn that into something measurable, and state the problem clearly enough that everyone in the room agrees on it.

How to run it — step by step

#	Instruction to learners	Facilitator note
1	Read the scenario and the raw feedback. Open data/voc-raw-customer-feedback.csv (or the Excel workbook) and read all 30 pieces of customer feedback — online reviews, comment cards and counter remarks. Underline every complaint you can find. Read before you write: the most common reason improvement projects fail is that the team decided what the problem was before looking at the evidence.	Watch for learners paraphrasing the VOC into business language — stop them. VOC stays in the customer's own words; the translation belongs in the CTQ column.
2	Capture five VOC statements in the customer's own words. Using templates/voc-to-ctq-template.csv, write down five Voice of the Customer statements quoted from the Verbatim Comment column. Keep the customer's own words — 'I'm late for work because the queue is too slow', not 'customers desire improved service efficiency'. The translation happens in the next step, not here.	Push back on any CTQ written without a unit ('serve faster'). Ask: how would you know next month whether this improved?
3	Turn each VOC into a measurable CTQ. For each VOC statement, write what the customer actually needs, then a Critical to Quality requirement that carries BOTH a number AND a unit — 'served within 5 minutes', not 'served faster'. If you cannot put a number on it, you cannot tell later whether you fixed it.	A problem statement that smuggles in a solution is the single most common White Belt error, and Case Study Q1 tests it directly. Read a few aloud and let the room spot the hidden solutions.
4	Write the problem statement. In templates/problem-statement-and-scope-template.csv, fill the four components: what is wrong, where and when it happens, since when, and how big it is — then the impact on the business. Write NO solution. 'Because we need to open a second till' is a solution and invalidates the statement.	If a table puts everything in scope, ask whether they could finish it in six weeks.
5	Test the goal against SMART. Rewrite your goal so it is Specific, Measurable, Achievable, Relevant and Time-bound. A complete goal names a metric, a baseline, a target and a date. Note the difference from the problem statement: the problem is what is wrong TODAY; the goal is what will	

	be true at the END.	
6	Agree the scope. Fill the in-scope / out-of-scope rows so the team knows where the project stops. Test each item with: could a White Belt team finish that in six weeks? Anything needing capital spend or a supplier renegotiation goes out of scope.	
7	Prepare the debrief line. Agree one sentence: "Our problem statement passes because it says _____, and it contains no solution." Nominate a spokesperson to read it out.	

Suggested timing

Minutes	Phase
4 min	Read the scenario and the 30 pieces of customer feedback
9 min	Extract VOC statements and convert each to a measurable CTQ
8 min	Draft the problem statement and test it against SMART
5 min	Agree what is in and out of scope
4 min	Prepare the debrief line

Total: 30 minutes

Discussion & decision prompts

1. Reading the raw feedback, what are customers actually asking for — and how does that differ from what the owner assumed the problem was?
2. Which of your CTQ requirements would be the hardest to measure in a real cafe, and how would you measure it anyway?
3. What makes a problem statement useful to an improvement team, and what makes one useless?
4. Why does a White Belt project need an explicit out-of-scope list?

Debrief

Bring the class back together and draw out the learning:

- Ask two or three groups to show what they produced, not just what they concluded.
- Ask what surprised them — the gap between what they assumed and what they found.
- Link the activity explicitly back to DMAIC · DEFINE — Define — Understand the Problem.
- Remind learners this activity rehearses what the Case Study assessment will require.

Reflection questions:

- Did your problem statement contain a hidden solution on the first attempt? Most do.
- Which piece of customer feedback surprised you most compared with what you expected to find?
- If the owner insists on buying the second coffee machine anyway, what would you say using only your VOC and CTQ table?

Facilitator notes — what to watch for

- Watch for learners paraphrasing the VOC into business language — stop them. VOC stays in the customer's own words; the translation belongs in the CTQ column.
- Push back on any CTQ written without a unit ('serve faster'). Ask: how would you know next month whether this improved?
- A problem statement that smuggles in a solution is the single most common White Belt error, and Case Study Q1 tests it directly. Read a few aloud and let the room spot the hidden solutions.
- If a table puts everything in scope, ask whether they could finish it in six weeks.

Success criterion

Every group can show five VOC statements in the customer's own words, five CTQs each carrying a number and a unit, and a problem statement containing no solution.

Assessment link

This activity builds DEFINE — Define — Understand the Problem. All 5 activities are in-class activities assessed indirectly through the open-book Case Study (CS) component, which reuses the same techniques on the same continuous BrewBean Cafe scenario. Knowledge is assessed in the Written Assessment (WA-SAQ). Nothing is assessed that is not practised here or taught in the slides.